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Structural Analysis of an Epidermal Growth Factor/Transforming Growth Factor-α Chimera with Unique ErbB Binding Specificity*

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Various chimeras of the ErbB1-specific ligands epidermal growth factor (EGF) and transforming growth factor-α (TGFα) display an enlarged repertoire as activators of ErbB2·ErbB3 heterodimers. Mutational analysis indicated that particularly residues in the N terminus and B-loop region of these ligands are involved in the broadened receptor specificity. In order to understand the receptor specificity of T1E, a chimeric ligand constructed by the introduction of the linear N-terminal region of TGFα into EGF, we determined in this study the solution structure and dynamics of T1E by multidimensional NMR analysis. Subsequently, we studied the structural characteristics of T1E binding to both ErbB1 and ErbB3 by superposition modeling of its structure on the known crystal structures of ErbB3 and liganded ErbB1 complexes. The results show that the overall structure of T1E in solution is very similar to that of native EGF and TGFα but that its N terminus shows an extended structure that is appropriately positioned to form a triple β-sheet with the large antiparallel β-sheet in the B-loop region. This conformational effect of the N terminus together with the large overall flexibility of T1E, as determined by ^{15}N NMR relaxation analysis, may be a facilitative property for its broad receptor specificity. The structural superposition models indicate that hydrophobic and electrostatic interactions of the N terminus and B-loop of T1E are particularly important for its binding to ErbB3.

The ErbB signaling network is composed of the ErbB1 (or EGFR), ErbB2 (HER2/Neu), ErbB3 (HER3), and ErbB4 (HER4) tyrosine kinase receptors. Upon ligand binding, these receptors dimerize into a variety of homodimeric and heterodimeric receptor complexes whereby the intrinsic kinases become activated, which results in a cascade of second messengers and a diversity of subsequent downstream signaling (1, 2). ErbB receptors play an important role in growth and differentiation of cells, whereas overexpression of both receptors and ligands has been found in several human cancers (3). As a consequence, the ErbB signaling network is increasingly used as a therapeutic target for the development of anti-tumor drugs (4, 5). Recent crystallographic studies have boosted this field of research by providing a wealth of information on the structure of ErbB-ligand complexes (6, 7).

A critical step in ErbB receptor signaling is the binding of EGF-like ligands to the extracellular domain of the receptor. More than a dozen soluble ligands have been identified that can be categorized into three distinct groups. A first group, composed of epidermal growth factor (EGF),[1] transforming growth factor-α (TGFα), and amphiregulin, binds specifically to ErbB1. A second group consists of neuregulin (NRG) with its multiple isoforms, which have specific affinity for ErbB3 and ErbB4. A third group binds to both ErbB1 and ErbB4 and is composed of β-cellulin, heparin-binding EGF, and epiregulin (8, 9). For ErbB2, no soluble ligand has been identified, but it forms the preferred dimerization partner for all other members of the ErbB family.

Despite having distinct receptor binding specificity, all ErbB growth factors have an EGF-like domain as a common motif, which is defined by three disulfide bridges that generate three looped regions, designated the A-, B-, and C-loop, in addition to linear N and C termini. A single hinge residue between the fourth and fifth cysteine divides the EGF-like ligand into an N- and C-terminal half, each with a two-stranded antiparallel β-sheet.

For several ligands, including EGF, TGFα, NRG-1α, and betacellulin, solution structures have been determined by NMR analysis under a variety of experimental conditions (10–15). Recently, also crystal structures have been reported of EGF (16), ErbB3 in unliganded form (17), and ErbB1 in complex with EGF (7) or TGFα (6). The last two structures established the relative orientation of the four domains in the extracellular part of the receptor and confirmed earlier observations that both EGF and TGFα bind a single ErbB1 receptor molecule, which subsequently dimerizes to form a 2:2 complex. Moreover, these studies indicated that specific residues in the B-loop region of EGF and TGFα are involved in binding to the first extracellular domain (domain I) of the ErbB1 receptor, whereas several residues in the A-loop, C-loop, and C terminus are in close contact with the third extracellular domain (domain III). Unlike EGF, also residues in the N terminus of TGFα directly contact the receptor in domain I.

Site-directed mutagenesis and phage display studies have

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The atomic coordinates and structure factors (code 1P9J) have been deposited in the Protein Data Bank, Research Collaboratory for Structural Bioinformatics, Rutgers University, New Brunswick, NJ (http://www.rcsb.org/).

The chemical shifts and ^{15}N relaxation data have been deposited in the BioMagnetic Resonance Bank under BMRB accession number 5801 (www.bmrb.wisc.edu/).

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[1] The abbreviations used are: EGF, epidermal growth factor; TGFα, transforming growth factor α; NRG, neuregulin; Mut, methanol utilization; HSQC, heteronuclear single-quantum coherence; NOE, nuclear Overhauser effect; TOCSY, total correlation spectroscopy; NOESY, nuclear Overhauser effect enhancement spectroscopy; r.m.s., root mean square.

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identified amino acids that have only limited freedom of mutation and are thought to be involved directly in receptor binding (18). For EGF and TGFα binding to ErbB1, these highly conserved residues are located around the second and sixth cysteine and in the C terminus (19, 20). In contrast, for NRG binding to ErbB3, hydrophobic and charged residues located in the N terminus and B-loop appear of primary importance (21–23). Supposing that all EGF-like ligands bind their respective ErbB receptor in a similar orientation, it appears that domain I of ErbB3 contributes mostly to ligand specificity, whereas for ErbB1 this is located in domain III (24, 25). In order to study ligand binding specificity, we have previously used a domain exchange strategy between EGF and TGFα and shown that replacing the N-terminal linear region of EGF with that from TGFα results in a chimera, designated here as T1E, with unique ErbB binding specificity. T1E not only maintains high affinity for ErbB1, the natural receptor for both EGF and TGFα, but has in addition gained the ability to bind to ErbB2·ErbB3 heterodimers with nearly identical affinity as NRG-1 (26). Subsequent mutation analysis showed that the TGFα residues His^4 and Phe^5 in the N terminus and the EGF residue Leu^{28} at the tip of the B-loop are essential for the high affinity of T1E for ErbB2·ErbB3 heterodimers. Replacement of Leu^{28} for the Glu residue present at the equivalent position in TGFα strongly impaired binding to ErbB2·ErbB3 heterodimers without affecting binding to ErbB1 (26). These results indicated that the combination of residues in the N terminus of TGFα and in the B-loop region of EGF mediate the enhanced binding affinity for ErbB2·ErbB3 heterodimers.

High affinity binding requires a proper three-dimensional structure of the ligand in combination with specific residues in the receptor binding domain. In order to understand the altered receptor specificity of T1E compared with EGF and TGFα, we have studied structure and dynamics of T1E by two- and three-dimensional heteronuclear NMR techniques. Subsequently, we modeled the interaction of T1E with both the ErbB1 and ErbB3 receptors, on the basis of the recently published crystal structures of the extracellular regions of ErbB3 (17) and the EGF·ErbB1 and TGFα·ErbB1 complexes (6, 7). Using these models, the specific roles of residues in the N terminus and B-loop of T1E in ErbB2·ErbB3 interaction were analyzed.

EXPERIMENTAL PROCEDURES

Construction of T1E Expression Vector—To produce sufficient amounts of T1E for NMR analysis, we used the *Pichia pastoris* protein expression system (Invitrogen). Thereto, T1E was cloned into the *P. pastoris* expression vector pPICZαA. A gene construct encoding the human TGFα/EGF chimera T1E, previously described by Stortelers *et al.* (26), was used as a template for the amplification of the T1E fragment by PCR. The primers used for PCR introduced an *Xho*I and a *Sal*I restriction site permitting directional cloning of the amplified DNA in frame with the α-factor leader sequence in the pPICZαA expression vector. A stop codon preceding the *Sal*I site prevented expression of the His tag. The amplified fragment was first subcloned into pCR2.1 TOPO® cloning vector (Invitrogen). The recombinant TOPO vector was digested with *Xho*I-*Sal*I to generate a 189-bp fragment, which was subsequently introduced into the pPICZαA expression vector using its *Xho*I-*Sal*I sites. The correctness of the constructs was confirmed by DNA sequencing. Approximately 7 μg of the DNA construct were linearized with *Bst*XI prior to transformation of *P. pastoris* X33 (Mut+) and KM71H (MutS) cells. The electroporation method of the EasySelect™ *Pichia* expression kit (version F; Invitrogen) was used for transformation, and the transformants were plated on MD and MM agar plates to screen for methanol utilization (Mut) phenotype. Several Mut+ and MutS clones were put on plates with very high zeocin concentrations (up to 1 mg/ml) to select for the so-called "jackpot" clones with multiple integrations. Southern blotting was used to confirm the presence of multiple integrations in the genome of selected high zeocin-resistant colonies. In small scale expression screens, the expression levels showed to be proportional to the number of integrations. A Mut+ clone with multiple integrations was selected for large scale expression and ^{15}N labeling.

Expression of Recombinant ^{15}N-labeled T1E in P. pastoris—To produce ^{15}N-labeled T1E, we slightly modified the protocol of Wiles *et al.* (27). In brief, 25 ml of BMG culture medium (0.1 M potassium phosphate (pH 6.0), 0.34% (w/v) yeast nitrogen base without amino acids or ammonium sulfate, 1% (v/v) glycerol, 0.00004% (w/v) biotin, and 1% (w/v) ammonium sulfate) were inoculated with a fresh colony of the selected Mut+ clone and grown overnight while shaking at 30 °C to midlogarithmic phase. The culture was scaled up to 500-ml batches of BMG medium without ammonium sulfate but containing 0.1% (w/v) ^{15}N-labeled ammonium chloride as its sole nitrogen source (99% ^{15}N; Campro Scientific, Veenendaal, The Netherlands) in 2-liter baffled flasks. After overnight culture, the cells were transferred to 500 ml of BMM induction medium (0.1 M potassium phosphate (pH 6.0), 0.34% (w/v) yeast nitrogen base without amino acids or ammonium sulfate, 0.5% (v/v) methanol, 0.00004% (w/v) biotin, and 1% (w/v) ^{15}N-labeled ammonium chloride) and grown for 9 days with additional supplies of methanol every 24 h. The minimal amount of nitrogen source needed for optimal protein production was determined to be 0.1% in the growth medium and 1% in the induction medium.

Protein Purification—The *P. pastoris* culture supernatant was subjected to cation exchange chromatography (SP-Toyopearl 550C, Tosoh Corp., Tokyo, Japan) and eluted with a linear salt gradient (0.2–1.0 M NaCl in 0.05 M NaAc). The fractions were tested for binding affinity to ErbB1 by an ^{125}I-mouse EGF binding competition assay as previously described by van de Poll *et al.* (28) and analyzed in parallel by nonreducing SDS-PAGE and Western blot analysis using a polyclonal antibody (Ab-3) raised against recombinant wild type human EGF (Oncogene Science Inc., Cambridge, MA). Specific antibody binding was detected by a goat anti-rabbit antibody linked to horseradish peroxidase and visualized by enhanced chemiluminescence. Fractions positive for ErbB1 binding and displaying only a single band on Western blot were collected, dialyzed against 0.2 M HAc, and lyophilized. After resolubilization in 0.5 M HAc, the protein was finally purified by reverse phase high pressure liquid chromatography on a Deltapak C_{18} column (Waters Associates, Milford, CT). Elution was carried out using a linear gradient of 20–40% acetonitrile in 0.1% trifluoroacetate. Using matrix-assisted laser desorption-ionization time-of-flight mass spectrometry, the purified protein was identified as ^{15}N-labeled T1E-(1–54) lacking the C-terminal arginine. The fractions corresponding to the main peak were collected, and the volume was reduced using speedvac centrifugation and subsequently dialyzed against double distilled water to remove remaining traces of organic solvents and finally against 50 mM phosphate buffer, pH 6.3. We thus obtained ~13 mg of uniformly ^{15}N-labeled T1E/liter of culture medium. The labeled material was biologically indistinguishable from the *E. coli*-derived T1E described previously (26).

Data Analysis and Structure Determination—The NMR sample was prepared to contain 0.8 mM protein in 95% H_2O, 5% D_2O (v/v), in 50 mM sodium phosphate buffer (pH 6.3) with 20 μg/ml Pefabloc (Roche Applied Science). NMR spectra were recorded at 298 K on Varian Unity Inova spectrometers operating at 800- and 600-MHz ^1H resonance frequencies. The following multidimensional experiments were recorded and analyzed: ^{15}N HSQC, HNHA, HNHB, two-dimensional TOCSY, two-dimensional NOESY, ^{15}N TOCSY, ^{15}N NOESY, and ^{15}N-filtered TOCSY (for both the aromatic and the aliphatic region). All spectra were processed on a Silicon Graphics work station using NMRPipe software (29). Backbone and side chain resonances were assigned using XEASY (30). Nuclear Overhauser effect (NOE) peaks from the two- and three-dimensional NOESY spectra were classified either as very weak, weak, medium, or strong with an upper distance restraint of 7.0, 5.0, 3.5, and 2.8 Å, respectively. Structure calculations were performed using the program XPLOR 3.851 (31). Dihedral angle restraints were derived from $^3J(H^NH)$-coupling constants, which were measured by a three-dimensional HNHA experiment (32). Subsequently, additional ϕ and ψ angle restraints were predicted using TALOS (33), resulting in a total of 98 dihedral angle restraints. To improve local geometry and electrostatics, the lowest energy structures were refined in water using a restrained molecular dynamics protocol under a CHARMM22 force field (34). The resulting ensemble of 36 structures was evaluated with MOLMOL (35), PROCHECK-NMR (36), and WHAT-IF (37). Figures were produced using the programs RasTop and MOLMOL. Structural data for comparison with the T1E structure were taken from Protein Data Bank entries 1YUG for the NMR structure of TGFα (14), 1HAF for the NMR structure of NRG-1α (13), 1JL9 for the crystal structure of EGF (16), and the Protein Data Bank files available on the World Wide Web at www.ocms.ox.ac.uk/idc/structures/egf (10) for the NMR structure of EGF. The crystal structures of ErbB3 and the

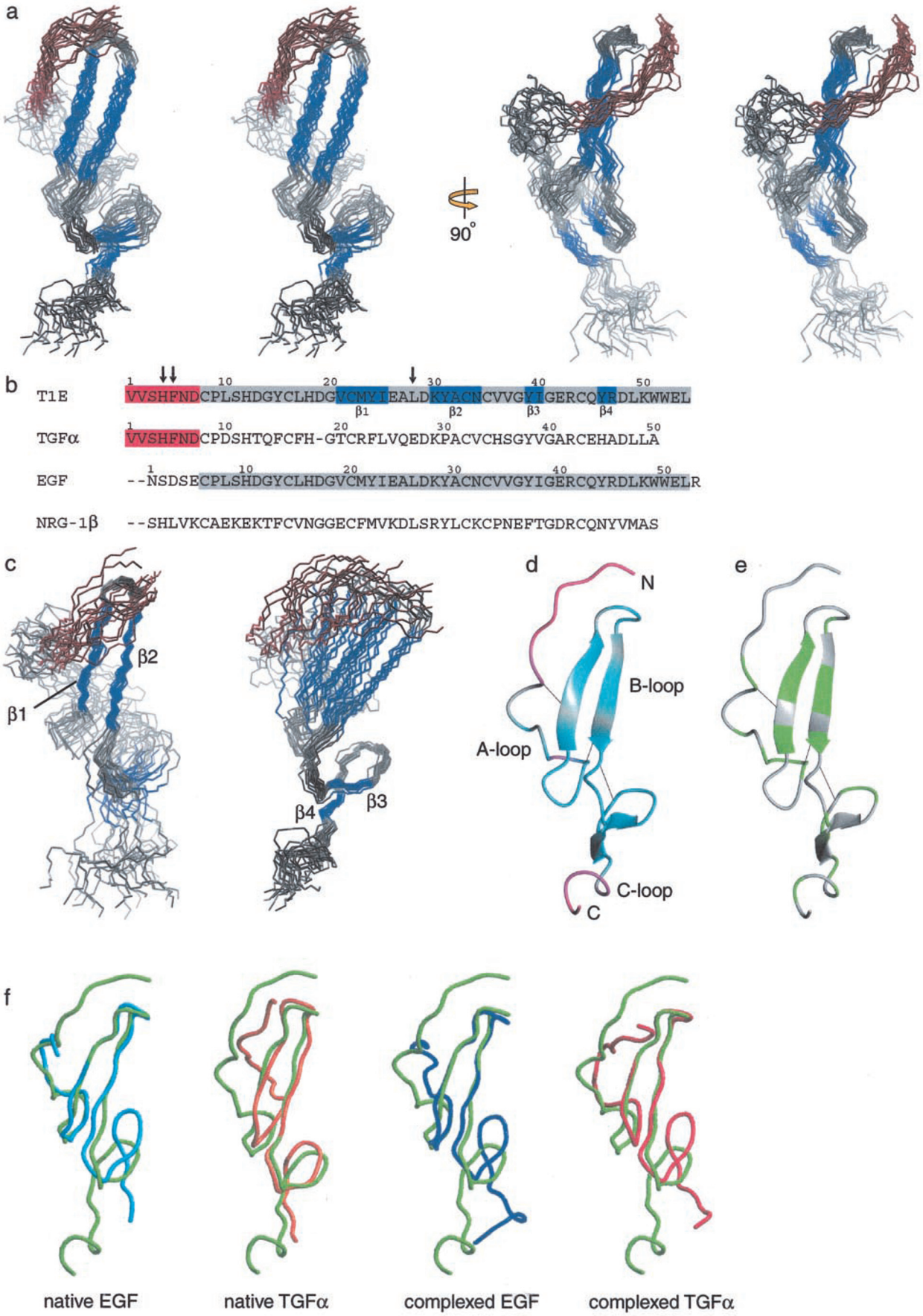

FIG. 1. **Solution structure of the EGF/TGFα chimera T1E.** *a*, stereo views of the ensemble of 36 structures of T1E in two orientations, superimposed using the backbone atoms of all residues (pairwise r.m.s. deviation 2.43 ± 0.63 Å). Residues *colored dark red* are based on TGFα; *gray* and *blue* (β-sheet)-*colored* residues are EGF-based. *b*, sequence alignment of T1E with TGFα, EGF, and NRG-1β. The *arrows* depict residues that mediate enhanced binding affinity for ErbB2·ErbB3 heterodimers. For clarity, the ω-loop of NRG-1β, which is known to be dispensable for

TABLE I
Structural statistics for the NMR structure of the EGF/TGFα chimera T1E

Structural statistics	T1E
Restraint information	
Distance restraints	660
(intraresidual/sequential/medium/long)	(309/194/67/90)
Hydrogen bonding restraints	9
Dihedral angle restraints	98
Disulfide bridge constraints	3
Atomic pairwise r.m.s. differences in backbone atoms (Å)	
Total protein	2.43 ± 0.63
B- and C-loop	1.61 ± 0.44
B-loop	1.27 ± 0.40
C-loop	0.91 ± 0.29
Ramachandran quality parameters (%)[a]	
Residues in most favored regions	82.9
Residues in additional allowed regions	16.4
Residues in generously allowed regions	0.7
Residues in disallowed regions	0
Average r.m.s. deviation (r.m.s. *Z*-scores, null deviation = 1)[a]	
Bond lengths	0.973
Bond angles	1.133
Omega angle restraints	1.502
Inside/outside distribution	1.153
Average deviation (*Z*-scores, null deviation = 0)[a]	
Coarse packing quality	−2.525
Ramachandran plot appearance	−2.097
χ-1/χ-2 rotamer normality	−2.470

[a] Values based on WHAT-CHECK output (53).

EGF·ErbB1 and TGFα·ErbB1 complexes were taken from Protein Data Bank entries 1M6B (17), 1IVO (7), and 1MOX (6), respectively.

Relaxation Measurements—^{15}N longitudinal relaxation rate R_1, the ^{15}N transverse relaxation rate $R_{1\rho}$, and the steady-state {^{1}H}-^{15}N NOE were measured at 298 K at 14.1 tesla field strength. R_1 relaxation rates were determined from a series of six interleaved spectra recorded with relaxation delays of 0.016, 0.128, 0.256, 0.512, 0.768, and 1.024 s. For the determination of $R_{1\rho}$ relaxation rates, six spectra were recorded in an interleaved manner with relaxation delays of 0.016, 0.032, 0.048, 0.064, 0.112, and 0.128 s. An overall rotational correlation time (τ_c) was first estimated from the mean ratio $R_{1\rho}/R_1$ and optimized to 5.4 ns using a grid search routine of the Modelfree program. This value is higher than expected for a monomeric 54-amino acid protein at 298 K and probably results from (partial) aggregation, since at concentrations above 0.8 mM the sample started to precipitate. Using the optimized τ_c value, the backbone dynamics were further analyzed on a per residue basis by Lipari-Szabo analysis (38, 39) using the program Modelfree 4.01 (40). We analyzed the generalized order parameter (S^2), which is correlated with backbone rigidity, the internal correlation time (τ_e), which is the time scale of internal motion (picosecond to nanosecond), and the chemical exchange contribution R_{ex} describing motions on millisecond to microsecond time scale.

Homology Modeling and Structural Superposition Modeling—To compare binding of T1E to the extracellular domains of ErbB1 and ErbB3, we built structural superposition models of T1E·ErbB1 domain I and T1E·ErbB3 domain I. Structures of T1E and ErbB1 domain I were available (1P9J (this paper) and 1MOX), but the crystal structure of ErbB3 (1M6B) published by Cho and Leahy (17) lacks the N-terminal 35 residues. To compensate for this lack of structural information, we first built a homology model of this region (residues 5–35) of ErbB3 by assuming that it has a similar structure as the corresponding region in ErbB1. In order to do so, sequences of ErbB1 domain I and ErbB3 domain I were aligned using ClustalW (41), which showed that this region of these two receptors has a sequence identity of 29.0% and a sequence similarity of 67.7%. The crystal structure of domain I of ErbB1 (1MOX) was subsequently used as a template structure for homology modeling with the program WHAT-IF (37).

RESULTS

Structure Calculation and Quality—Using a series of homonuclear and heteronuclear NMR experiments, ^{1}H and ^{15}N resonance assignments were obtained for T1E using the Wüthrich sequential assignment protocol (42). The proton resonance assignments were mainly derived from two-dimensional TOCSY and two-dimensional NOESY spectra, whereas a three-dimensional ^{15}N NOESY spectrum was used to identify the sequential ^{15}N resonances of backbone amide groups and to verify the obtained proton resonance assignments.

From the NOE spectra, a set of 660 NOEs was collected, composed of 309 intraresidual, 194 sequential, 67 medium range, and 90 long range NOEs. This resulted in an average of 12.2 distance restraints per residue. Additional constraints included nine hydrogen bonding restraints, three constraints from the disulfide bridges, and 98 ϕ and ψ angle restraints. We calculated an ensemble of 36 high resolution NMR structures of T1E. A superposition of this ensemble in two different orientations is shown in stereo view in Fig. 1*a*. The structural statistics are listed in Table I. All accepted structures contained neither distance violations greater than 0.5 Å nor angle violations greater than 5° from experimental data. Analysis of the Ramachandran plot showed that for the 36-structure ensemble 82.9% of the residues were found in the most favored regions, 16.4% in the additionally allowed regions, and 0.7% in generously allowed regions. No residues were found in the disallowed regions of the Ramachandran plot.

The data presented in Fig. 1*a* show that the protein contains two major stretches of secondary structure, consistent with the EGF fold. A large anti-parallel β-sheet is formed by residues Val^{21}–Ile^{25} (β_1) and Lys^{30}–Asn^{34} (β_2), and a smaller anti-par-

ErbB3 binding, is not shown. *Color coding* is as in *a*. *c*, T1E structure ensemble superimposed for B-loop residues (pairwise r.m.s. deviation 1.27 ± 0.4 Å) and C-loop residues (pairwise r.m.s. deviation 0.91 ± 0.29 Å), respectively. *d*, *ribbon* presentation of the T1E structure *color-coded* according to residue-specific S^2 values. Residues with S^2 = 0–0.7 are displayed in *pink*, S^2 = 0.7–0.8 in *violet*, and S^2 = 0.8–1.0 in *light blue*. No data are available for residues *colored gray*. The A-, B-, and C-loop as well as N and C termini are indicated, and the cysteine bridges are depicted by *gray lines*. *e*, *ribbon* presentation of the T1E structure *color-coded* according to residue-specific R_{ex} values. Residues requiring a chemical exchange contribution are shown in *green*. *f*, superposition of the lowest energy structure of T1E onto EGF and TGFα in native and complexed forms. T1E is shown in *green*, native EGF in *light blue* (molecule A from 1JL9), native TGFα in *orange* (1YUG, mean structure), EGF complexed to ErbB1 in *blue* (from 1IVO.pdb), and TGFα complexed to ErbB1 in *red* (from 1MOX.pdb).

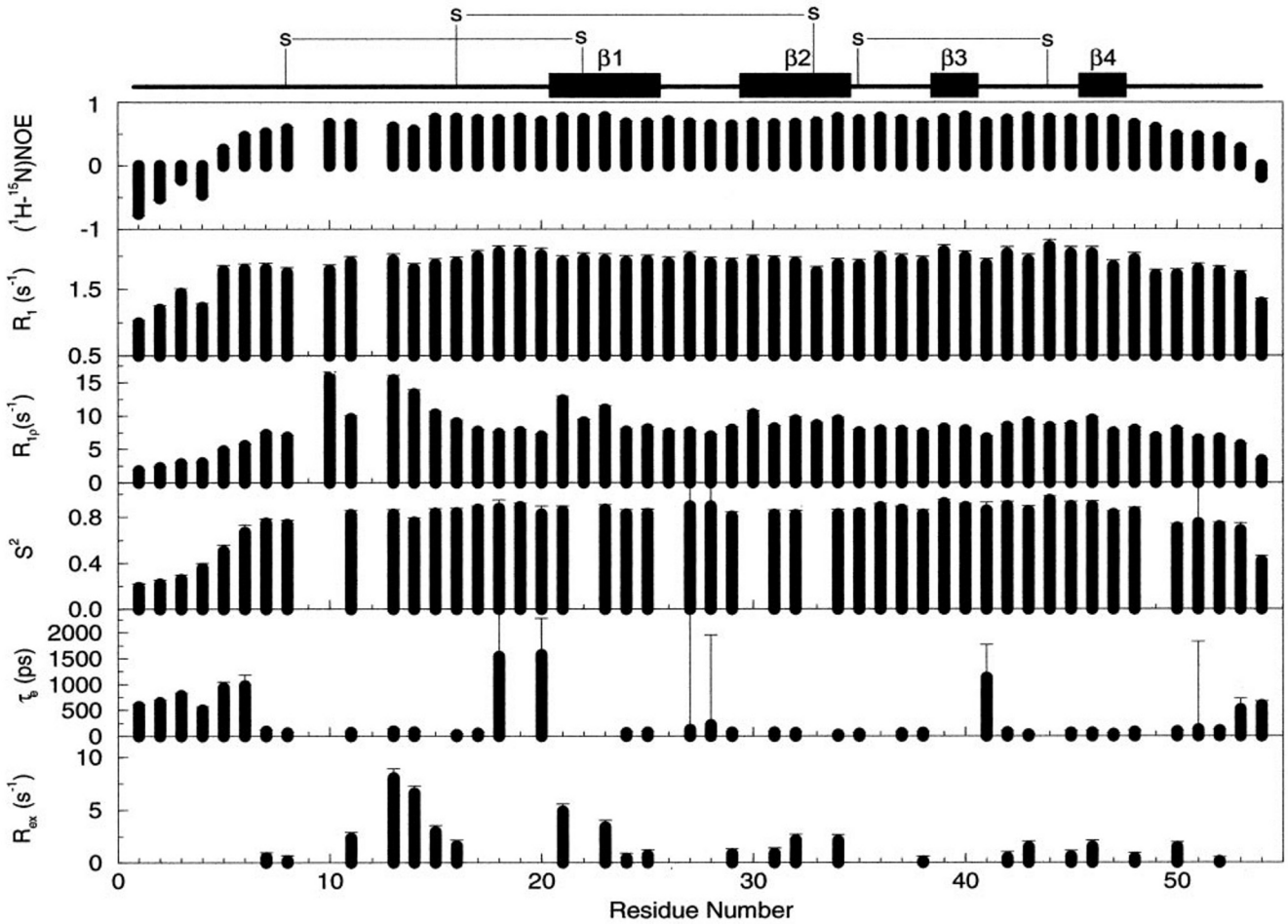

FIG. 2. **Dynamics of the EGF/TGFα chimera T1E.** Shown is a plot of the ^{15}N relaxation data (R_1, $R_{1\rho}$, and NOE enhancement), and subsequent Modelfree analysis data (S^2 generalized order parameter, τ_e internal correlation time, and R_{ex} chemical exchange contribution) of T1E as a function of residue number. The β-sheets in the secondary structure of T1E and location of cysteine bridges are indicated at the *top*.

allel β-sheet is formed by residues Tyr^{39}–Ile^{40} (β_3) and Tyr^{46}–Arg^{47} (β_4) (see Fig. 1*b*). These two structural elements are connected by three disulfide bridges, located between Cys^8 and Cys^{22}, Cys^{16} and Cys^{33}, and Cys^{35} and Cys^{44}. In Fig. 1*c*, the T1E structures in the ensemble were superimposed for B-loop residues (*left panel*) and C-loop residues (*right panel*), revealing good convergence to a single fold in the regions containing the large and small β-sheets (in *blue*), with a global pairwise backbone r.m.s. deviation of 1.27 ± 0.4 Å and 0.91 ± 0.29 Å for B- and C-loop, respectively. The relative orientation of the N-terminal and C-terminal halves is determined primarily by 25 long range distance constraints derived from NOEs between N-terminal half residues (Leu^{17}, His^{18}, Gly^{20}, Val^{21}, or Met^{23}) and C-terminal half residues (Asn^{34}, Val^{36}, Arg^{43}, Cys^{44}, or Gln^{45}). Due to the elongated shape of the molecule, these NOEs cluster in a limited region at the base of the large β-sheet, and consequently no structural information is available connecting the B-loop region directly to the C-loop region. As a result, the pairwise r.m.s. deviation for the entire protein is inflated to 2.43 ± 0.63 Å.

Notably, the N terminus adopts an extended conformation that approximately follows the β_1-strand, although it is much less ordered than the large β-sheet in the B-loop itself (Fig. 1*a*). Its conformation is determined, however, by several unambiguous, long range NOEs between Val^1 and Leu^{28}, Val^2 and Leu^{28}, Ser^3 and Leu^{28}, Phe^5 and Met^{23}, and Asn^6 and Tyr^{24}. In 4 of 36 structures in the ensemble, this resulted in a triple β-sheet formation of the N terminus with the antiparallel β-sheet in the B-loop, based upon criteria used by PROCHECK-NMR.

TABLE II
Residues that are involved in N terminus and B-loop interactions of EGF and TGFα with ErbB1 (6, 7) and equivalent residues in ErbB3 based on structural sequence alignment

TGFα residues	ErbB1 residues	Corresponding residues in ErbB3
His^4	Tyr^{101}	Tyr^{104}
Phe^5	Tyr^{45}	Leu^{48}
Arg^{22}	Tyr^{45}	Leu^{48}
Gln^{26}	Tyr^{101}	Tyr^{104}
	Asn^{128}	Lys^{132}
Glu^{27}	Arg^{125}	Tyr^{129}
	(Tyr^{89})	Tyr^{92}
Lys^{29}	Glu^{90}	Asp^{93}
EGF residues	**ErbB1 residues**	**Corresponding residues in ErbB3**
Met^{21}	Tyr^{45}	Leu^{48}
Ile^{23}	Leu^{14}	Leu^{17}
	Leu^{69}	Met^{72}
Leu^{26}	Leu^{98}	Met^{101}
Lys^{28}	Glu^{90}	Asp^{93}

Relaxation and Dynamics Parameters—Despite the high degree of structural similarity, most members of the EGF family of polypeptide growth factors show marked differences in receptor binding specificities. Since ligand binding may be influenced by intrinsic dynamical properties of the protein, we characterized the backbone dynamics of T1E.

The nice dispersion in the ^{15}N HSQC spectra allowed the measurement of {^{1}H}-^{15}N NOE values for all but one nonproline residue. Data for His^{12} did not yield reliable fits in the relaxation analysis. As shown in Fig. 2, {^{1}H}-^{15}N NOE values averaged between 0.7 and 0.8 for residues 15–23 and 34–47, indicating relatively little backbone motions on a picosecond to nanosecond time scale. Adjacent regions, such as the tip of the B-loop, had {^{1}H}-^{15}N NOE values between 0.6 and 0.7, indicating more backbone mobility. The N- and C-terminal ends of T1E revealed extensive local flexibility on a fast time scale indicated by NOE values ranging from 0.6 down to −0.755.

To further analyze the backbone internal motions, the Modelfree approach was used (38, 39). Backbone amide order parameters S^2 were determined for 46 residues (Fig. 2). These S^2 values were grouped into three classes, which are visualized in Fig. 1*d* using a *color-coding* scheme. The local flexibility of the N- and C-terminal ends is reflected by S^2 values of 0.2–0.7 and τ_e values of >500 ps. Interestingly, the residues in the N and C termini, as well as His^{18}, Gly^{20}, Ala^{27}, Leu^{28}, Gly^{41}, and Trp^{51}, could only be fitted using a model incorporating an intermediate time scale motional parameter, although for the nonterminal residues the fit yielded relatively large *error bars* on the accompanying τ_e values. In Fig. 1*e*, a *ribbon* presentation of the lowest energy T1E structure is shown with residues that required a contribution of the transverse relaxation rate R_{ex} in *green*. Non-zero values of R_{ex} were found for several residues mainly in the A-loop (residues Ser^{11}–Cys^{16}) and to a lesser extent also in the B-loop and C-loop region, which is indicative for chemical exchange processes within the millisecond to microsecond time range (Fig. 1*e*). Overall, the structure of T1E is relatively flexible throughout the entire amino acid sequence, with enhanced backbone flexibility and internal motions in the N and C termini and chemical exchange contributions in A-, B-, and C-loop regions.

Structural Superposition of T1E with Related Structures—Since the chimera T1E is composed of parts of the EGF and TGFα sequences, we compared the T1E structure with the parental molecules by superposition. We superimposed the lowest energy T1E structure onto native EGF (molecule A from crystal dimer 1JL9) and native TGFα (mean NMR structure 1YUG) as well as onto crystal structures of EGF and TGFα in the receptor-bound state, as obtained from Protein Data Bank entries 1IVO and 1MOX, respectively (Fig. 1*f*). Differences in relative orientations of the N- and C-terminal halves resulted in relatively high overall r.m.s. deviation values of 5.83, 4.22, 4.78, and 3.62 Å for superposition onto native EGF, native TGFα, complexed EGF, and complexed TGFα, respectively. However, the B-loop region of T1E superimposed very well onto the B-loop region of both receptor-bound EGF and TGFα, with r.m.s. deviation values of only 0.92 and 0.86 Å, respectively.

A Comparison of T1E Binding to ErbB1 Domain I and ErbB3 Domain I—T1E is unique in its properties since it can bind both to ErbB1 and ErbB3. Since the amino acid requirements for ligand binding to ErbB1 and ErbB3 are very much different, the three-dimensional structure of T1E provides the opportunity to compare the structural requirements for binding of a single ligand to two distinct receptor molecules. Domain exchange and phage display studies of T1E (26, 43) indicated that residues His^{4} and Phe^{5} in the N terminus and Leu^{28} in the B-loop region of T1E determine its affinity for ErbB2·ErbB3 binding, strongly suggesting a direct role for the involvement of receptor domain I. This is further substantiated by the observation that these three residues (see the *arrows* in Fig. 1*b*) are functionally conserved in NRG-1β, the natural ErbB2·ErbB3 agonist. Table II lists the amino acids in EGF and TGFα that directly interact in the crystal structures with specific residues in domain I of ErbB1 (6, 7) and also surveys the corresponding residues in domain I of ErbB3. Most residues in ErbB1 involved in ligand binding are conserved if not identical in ErbB3. Examples are Tyr^{101}, which is known to interact with His^{4} in the free N terminus of TGFα, and Glu^{90} which is responsible for the formation of a salt bridge with Lys^{28} in the β_2-strand of the B-loop of EGF (Lys^{29} in TGFα). Furthermore, Tyr^{45} and three Leu residues in ErbB1 that interact with B-loop residues of EGF have a conserved hydrophobic character. In contrast, Asn^{128} and Arg^{125}, two residues in ErbB1 that interact with residues at the tip of the B-loop of TGFα, are clearly different in ErbB3, comprising residues Lys^{132} and Tyr^{129}, respectively.

To visualize possible critical interactions in ligand-receptor binding, we generated structural models by superimposing the lowest energy T1E structure onto the domain I regions of both ErbB1 and ErbB3 receptor. Focus was laid on the ligand-binding interface of domain I, since this domain is most probably involved in binding the N-terminal and B-loop region of T1E that confer specificity toward ErbB2·ErbB3 complexes (26). The structural superposition modeling was composed of a three-step procedure. We first superimposed T1E onto complexed TGFα. The structure of complexed TGFα was preferred over that of complexed EGF, because only the N terminus of TGFα is known to be involved in receptor binding (6). In the second step, the complete domain I of ErbB3 (with the homology-modeled N-terminal region; see "Experimental Procedures") was superimposed on ErbB1 domain I obtained from the crystal structure of TGFα·ErbB1. Finally, an overlay of these two sets was made based on the orientation of complexed TGFα in both superpositions, thereby assuming that T1E binds ErbB1 in a similar manner as TGFα does and that binding to ErbB3 domain I is similar to binding to ErbB1 domain I. Fig. 3*a* compares the hydrophobic character of the receptor surface residues in these structural superposition models of T1E·ErbB1 domain I and T1E·ErbB3 domain I. Of particular interest are the residues juxtaposed to the crucial N-terminal T1E residues His^{4} and Phe^{5}. In the T1E·ErbB1 complex, His^{4} is in close proximity to Ser^{99} and Tyr^{101}, allowing for stacking interaction with the latter. This interaction appears to be preserved in the T1E·ErbB3 complex, where the homologous Tyr^{104} replaces Tyr^{101}, whereas Leu^{102} occupies the position of Ser^{99}. Similarly as in TGFα, T1E-Phe^{5} is oriented toward Leu^{69} and Tyr^{45} in the T1E·ErbB1 complex. This residue faces a much more hydrophobic environment in the T1E·ErbB3 complex, formed by Met^{72} and Leu^{48}. Furthermore, ErbB1 contains a large Phe residue at position 20. In ErbB3, two small residues, Gly^{50} and Ala^{23}, are located in the corresponding region, making the hydrophobic cavity in ErbB3 larger and better available for large hydrophobic N-terminal ligand residues. These differences in hydrophobicity between ErbB1 domain I and ErbB3 domain I strongly suggest a role for specific hydrophobic contacts in ligand-ErbB receptor binding selectivity.

In addition to specific hydrophobic contacts, we compared the contribution of electrostatic interactions in receptor binding specificity. Fig. 3*b* displays the electrostatic potentials for the ErbB1 domain I and ErbB3 domain I complexed with T1E. The ligand binding interface of the ErbB3 domain I shows a larger surface with electronegative potential than the corresponding surface of ErbB1 domain I. Residues jointly responsible for this electronegative patch on ErbB3 domain I are Glu^{45}, Asp^{93}, and Glu^{131}. Furthermore, the positive electrostatic potential resulting from Arg^{125} and Lys^{13} in ErbB1 are absent in ErbB3. In Fig. 3*c*, the receptor binding interface of the ligands is shown. T1E, EGF, and TGFα are displayed as a surface representation with electrostatic potentials colored as in Fig. 3*b*, together with the backbone trace of ErbB1 domain I. The models in Fig. 3*c*

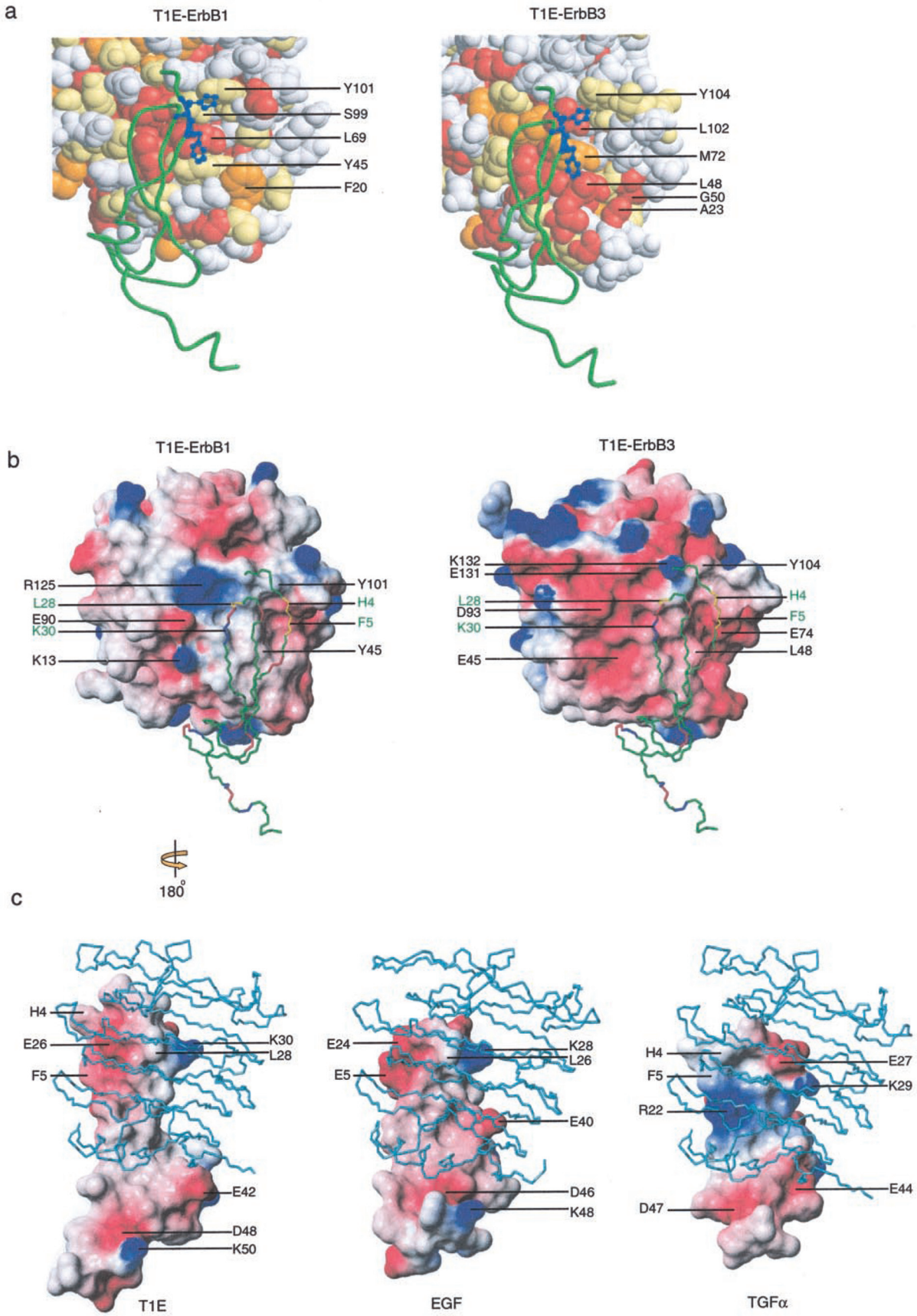

FIG. 3. **Comparison of the T1E·ErbB1 domain I and T1E·ErbB3 domain I complexes obtained by structural superposition.** *a*, hydrophobic interactions of T1E N terminus with domain I of ErbB1 and ErbB3. The backbone trace of T1E is colored *green* with selected side chains His[4] and Phe[5] shown in *ball-and-stick* representation (*blue*). Domain I of both receptors is shown in a *space-filling* representation and *colored* according to hydrophobicity (52), ranging from *yellow* (less hydrophobic) to *red* (more hydrophobic). *b*, electrostatic potential plots of ligand-binding interfaces of ErbB1 domain I and ErbB3 domain I in surface representation. Relative orientation of T1E is shown by the green backbone trace. Backbones of residues His[4], Phe[5], and Leu[28] are shown in *yellow*. Backbone traces of acidic and basic residues of T1E are *colored*

are rotated 180° around the *vertical axis* with respect to Fig. 3*b*. In the TGFα·ErbB1 complex, Glu^{27} at the B-loop tip of TGFα interacts with Arg^{125} of ErbB1 (6). In comparison, this negatively charged Glu^{27} of TGFα would probably be located in an unfavorable negatively charged environment on ErbB3, in particular close to Glu^{131}. In line with these considerations, previous work showed that in EGF/TGFα chimeras, Glu^{27} prevents interaction with ErbB2·ErbB3 complexes, whereas the corresponding Leu in EGF is favorable to this interaction (26). Furthermore, a comparison of the electrostatic potentials of the three ligands (Fig. 3*c*) reveals that Arg^{22} in TGFα is responsible for a large positively charged patch, whereas EGF is more negatively charged on this side of the molecule as a result of the presence of Glu^{5} and Glu^{24} in this region. T1E lacks Glu^{5}, but Glu^{26} at the end of the β_1-strand is also responsible for a small negative patch on T1E. Together, these structural superposition models show large differences in electrostatic potentials of both ErbB receptors and EGF-like ligands, indicating that electrostatic attraction and repulsion play a major role in ErbB receptor specificity of ligand binding.

DISCUSSION

EGF, TGFα, and the chimera T1E all bind to the ErbB1 receptor with high affinity. Of these three ligands, T1E is unique because it is also able to interact with ErbB2·ErbB3 heterodimers (26). Ligand binding results from a proper conformation in combination with the presence of specific residues at positions that directly interact with the receptor. In the present study, we have determined the solution structure and quantified the backbone dynamics of T1E. To explain the binding selectivity of T1E, we have constructed models of T1E·ErbB1 domain I and T1E·ErbB3 domain I, using structural superposition. The present data indicate that the additional affinity of T1E for ErbB2·ErbB3 heterodimers most probably stems from the interplay of three different factors: (i) the conformational effect of the N terminus, (ii) the dynamic properties of T1E, and (iii) the presence of specific residues that form part of the binding epitope for domain I of ErbB3. The structural superposition models of T1E·ErbB1 domain I and T1E·ErbB3 domain I strongly suggest that both hydrophobic interactions and electrostatic attraction/repulsion may be prominent aspects in determining ErbB ligand binding selectivity. In addition, our data provide structural support for the observation that His^{4} and Phe^{5} in the N terminus and Leu^{28} in the B-loop region are critical residues in mediating ErbB3 binding affinity.

The overall structural fold of T1E in solution is very similar to the solution structures of other members of the EGF family, such as EGF and TGFα (10, 14): an N-terminal half with a flexible loop region (A-loop) and a large antiparallel β-sheet (forming the B-loop) and a C-terminal half that contains a minor antiparallel β-sheet (forming the C-loop). The N- and the C-terminal halves of EGF-like ligands are connected by a hinge residue, Asn^{34} in the case of T1E. In comparison with EGF and TGFα, the N terminus of T1E displays ϕ,ψ-angles consistent with an extended conformation. In 4 of 36 structures in the ensemble, the N terminus explicitly forms a triple β-sheet with the B-loop. The observation that NRG-1α, which is a natural ligand for the ErbB2·ErbB3 heterodimer, also contains a triple β-sheet in this structurally homogeneous region (13) suggests that stabilization of the N-terminal region into an extended structure could be a requirement for EGF-like growth factors to bind ErbB3. Although the N terminus shows enhanced backbone flexibility, receptor binding of suitably folded T1E may drive the equilibrium of T1E into the receptor-bound, triple β-sheet-containing structure.

Whereas the secondary structure and global fold of EGF-like structures are quite similar, the overall r.m.s. deviations of the superpositions are relatively high (Fig. 1*f*). This may represent real differences between these structures or, alternatively, suggest that some of the structures are less well defined, either too loose or too tight. In our study, we used a CHARMM22 water refinement protocol (34) to improve the quality of the structure ensemble. This resulted in reliable *Z*-scores for the local geometry (Table I). Furthermore, the Ramachandran *Z*-score, which is a good indicator of the overall quality of the ensembles, yielded highly acceptable values for NMR structure ensembles. Hence, we consider our structure to be reliable in terms of a representation of our experimental data.

Although uncertainty in NMR data does not necessarily indicate motion between the N- and C-terminal halves, the presence of hinge-bending motions has previously been suggested for ErbB ligands (14, 44–47). The present observation that the B- and C-loop regions of the T1E ensemble superimpose nicely (see Table I) suggests that such motions could indeed be at play. In addition, many residues in T1E (Asp^{13}, Gly^{14}, Tyr^{15}, Cys^{16}, Val^{21}, Glu^{42}, Arg^{43}, Gln^{45}, and Tyr^{46}) at the interface between the N- and C-terminal halves exhibit nitrogen-15 exchange line broadening (Fig. 1*e*), similarly as observed previously for TGFα, which is consistent with the existence of hinge-bending motions. Furthermore, superposition of TGFα structures in bound and native state revealed a change in relative orientation of the N- and C-terminal halves, which could indicate that bending of the hinge is a prerequisite for proper receptor binding (6).

Dynamics may play a significant role in broad receptor binding specificity of T1E. From the backbone relaxation data of T1E and the subsequent Modelfree analysis, it appears that T1E is a relatively flexible molecule. The protein exhibits significant backbone motions in the picosecond to nanosecond time scale, and ~40% of the residues require a chemical exchange contribution. The residues with the largest R_{ex} values are located in the A-loop (Figs. 1*e* and 2). Furthermore, the data show R_{ex} values for groups of residues that are separated in sequence but close together in space (Asp^{7} and Cys^{8} (in the N terminus); Met^{23}, Tyr^{24}, and Ile^{25} (in the β_1-strand); and Tyr^{31} and Ala^{32} (in the β_2-strand)) that may be indicative of synchronous motion of these domains. It has been proposed that flexibility may correlate with broadened receptor-ligand binding specificity (48), and the observed flexibility of T1E may thus be a facilitative property for its broad receptor specificity. T1E can adopt multiple conformations, thus facilitating binding to domains I and III of both ErbB1 and ErbB3 receptors by an induced fit mechanism. In this view, the conformation of T1E in which the N terminus and the B-loop form a triple β-sheet could particularly be favorable for binding to domain I of ErbB3. If only 10% of the total available T1E is in the correct conformation to bind ErbB3, as indicated by the percentage of structures showing a triple β-sheet, this could explain the

red and *blue*, respectively. Residues potentially involved in specific ligand-receptor interactions are indicated by *number* (T1E residues in *green*). *Dark red* indicates electronegative potential, *dark blue* indicates positive electrostatic potential, and *white* indicates a potential of zero. Electrostatic potentials were calculated using the program MOLMOL (35). *c*, electrostatic potential plots of receptor-binding interfaces of T1E, EGF, and TGFα in surface representation. Relative orientation of the ErbB1 domain I is shown as a *light blue* backbone trace. For the T1E·ErbB1 complex, the structural superposition model is shown, whereas for the EGF·ErbB1 and TGFα·ErbB1 complex, the crystal structures with Protein Data Bank entry 1IVO (7) and 1MOX (6) were used, respectively. The orientation of the models is rotated 180° around the *vertical axis* with respect to *b*. Electrostatic potential *color coding* is as in *b*.

relatively low affinity of T1E for ErbB3 receptors alone and the need for stabilization by ErbB2 (26). A role for dynamics in the ligand-receptor recognition process has previously also been suggested for NRG-1α (49), where flexibility was observed in regions that are clearly important for receptor binding, such as both the N- and C-terminal regions (23). Our relaxation data, together with those described for other ErbB ligands, indeed suggest that dynamic flexibility contributes to broad specificity of the ErbB receptor recognition process.

To elucidate the role of specific residues in N terminus and B-loop of T1E for ErbB3 receptor binding, we made structural superposition models of T1E·ErbB1 domain I and T1E·ErbB3 domain I. From these models, we propose that the N terminus of T1E is involved in binding to ErbB3 by hydrophobic packing. His^4 and Phe^5, two residues in the N terminus of T1E that were earlier found to be important for ErbB2·ErbB3 binding specificity (26), are in proximity to a hydrophobic pocket in ErbB3 (Fig. 3*a*). The observation that this pocket is more hydrophobic and larger than the corresponding region in ErbB1 suggests that it favors binding of large hydrophobic residues such as His^4 and Phe^5. Moreover, the preformed extended conformation of the N terminus in T1E could facilitate surface presentation of His^4 and Phe^5, enhancing the formation of the required intermolecular contacts. In addition, these large hydrophobic residues in the N terminus of T1E could also be involved in compensating for less favorable acidic residues in the B-loop region. Unfavorable electrostatic interactions between Glu^{26} in T1E and Glu^{74} in ErbB3 could potentially be masked by His^4 and Phe^5 in the N terminus. This hypothesis is supported by phage display studies, which indicated that for optimal ErbB2·ErbB3 binding EGF should preferentially contain an aromatic, a hydrophobic, and a positively charged amino acid in its linear N terminus (43). The observation that a positively charged residue is favored in one of the positions in the N terminus suggests that masking the acidic Glu residue present at the end of the β_1-strand (corresponding to Glu^{26} of T1E) is required for proper binding to the ErbB3 domain I.

Several studies with mutant growth factors have suggested that acidic residues in the B-loop region are involved in ErbB binding selectivity. Chimeras that contain the B-loop region of TGFα are unable to activate ErbB2·ErbB3 heterodimers (26, 50), particularly because the acidic Glu^{27} at the tip of the TGFα B-loop is a negative determinant for interaction with ErbB2·ErbB3 heterodimers (26). In the present T1E·ErbB3 domain I model, Leu^{28} is positioned at the tip of the B-loop of T1E pointing toward a large electronegative patch on the ErbB3 ligand binding site, whereas the corresponding region in ErbB1 is predominantly positively charged. In the TGFα·ErbB1 receptor complex, the corresponding Glu^{27} is involved in an electrostatic interaction with Arg^{125}, thereby stabilizing the ligand-receptor complex (6). In contrast, the equivalent region in ErbB3 does not contain positively charged residues that could stabilize the interaction, as shown in Fig. 3*b*. We therefore argue that the large electronegative patch, in particular Glu^{131} in domain I of ErbB3, is responsible for electrostatic repulsion of Glu^{27} at the tip of the B-loop of TGFα, thereby inhibiting this interaction.

In the present study, we determined the solution structure and dynamics of T1E and analyzed its unique ErbB receptor binding specificity by using structural superposition models of T1E·ErbB1 domain I and T1E·ErbB3 domain I. Our results provide new insights into the subtle mechanism of ligand binding specificity of the ErbB receptor family. An attractive goal is to combine residues specific for ErbB1 and ErbB3 binding into one chimeric EGF-like molecule. Recent studies from our laboratory have shown that the affinity of T1E for ErbB3 binding can be enhanced significantly by mutations in both the N- and C-terminal regions (43, 51), without loss of ErbB1-specific affinity. By using a chimera with a combination of these mutations, a direct comparison of ErbB1 and ErbB3 binding mechanisms can be studied on the basis of a single ligand.

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